Lesson 1.13-1.14: Function Models and Applications

Topic 1.13: Function Model Selection and Assumption Articulation
Topic 1.14: Function Model Construction and Application

This lesson will help us to construct linear, quadratic, cubic, or piecewise-defined function models and applications given a real-world scenario.

How to Identify Function Types

Linear

- ❑ The output values will always increase or always decrease at (roughly) a constant rate of change.

Quadratic

- ❑ Context that involves area can be modeled by a quadratic function.
- ❑ Data sets will be (roughly) symmetric with a unique maximum or minimum value.

Cubic

- ❑ Geometric contexts involving volume or three dimensions.

Piecewise

- ❑ A data set defined over non-overlapping domain intervals.

EX #1: In each data sets below, give an explanation to justify why the model fits.

A. The data set models a linear function.

The rate of change is constant

x	-3	-2	-1	0	1
$f(x)$	11	7	3	-1	-5

-4 -4 -4 -4

B. The data set models a quadratic function.

The 2nd differences are constant

or the rate of change is linear

x	-1	0	1	2	3
$g(x)$	15	5	-1	-3	-1

-10 -6 -2 $+2$

$+4$ $+4$ $+4$

© 2026 Flamingo Math.com

EX #2: The data set for a piecewise-defined function h is given in the table below.

x	$h(x)$	
−3	−13	+3
−2	−10	+3
−1	−7	+3
0	−4	−1.5
1	−5.5	−1.5
2	−7	−1.5
3	−8.5	

A. Determine the two open intervals where the rates of change are two different constants.

$(-3, 0)$ $(0, 3)$

B. Write the piecewise function that models the data.

$$h(x) = \begin{cases} 3x - 4, & -3 < x < 0 \\ -1.5x - 4, & 0 < x < 3 \end{cases}$$

Geometric Models

Let's use a graphing utility to draw a scatter plot then create a line of best fit to analyze a data set. (*Note: Instructions for TI-84 or TI-Nspire included.*)

Fresh Market Price of Florida Oranges in Dollars/Box							
Year	2006	2007	2008	2009	2010	2011	2012
Price per Box	$8.40	$15.99	$10.75	$8.80	$12.75	$12.91	$13.12

1 2 3 4 5 6 7 8 9 10
2013 2014 2015

EX #3: The fresh market price of Florida oranges in dollars per box is shown in the table above.

A. Write the linear regression model for the price of oranges in dollars per box. $y = 0.357x + 10.389$

Stat → #1 edit Stat → calc → #4 alpha F4, #1

Enter data in L1 and L2

L1	L2	L3 2
1	8.4	------
2	15.99	
3	10.75	
4	8.8	
5	12.75	
6	12.91	
7	13.12	

Select 4 or 8: LinReg

LinRes
y=ax+b
a=.3571428571
b=10.38857143

Enter L1, L2 and Y1

LinReg(ax+b)
Xlist:L1
Ylist:L2
FreqList:
Store RegEQ:Y1
Calculate

Enter twice
(or hit zoom if using an Evo)

Calculate the equation

LinRes
y=ax+b
a=.3571428571
b=10.38857143

B. Use your linear function to estimate the price for a box of oranges in 2014 and 2015.

2014 : $f(9) = \$13.60$

2015: $f(10) = \$13.96$

- Turn Plot 1 on
- Hit Zoom 9 to see scatterplot & regression function
- Edit Window → X-max: 11
- Trace ↑ 9 and Trace ↑ 10

	0	1	2	3	4	5	6
time, t	Jan	Feb	Mar	April	May	June	July
Length of call, L	8.7	9.2	10.1	11.2	14.6	17.5	

EX #4: The average length of cell phone calls, L in minutes, for a high school student during the second semester (January to June) are shown in the table above. The model is known to be valid for January to September.

A quadratic regression $y = ax^2 + bx + c$ can be used to model these data, where L is the average length of a call and $t = 0$ represents January.

A. Write the equation of the quadratic model for these data.

$L(t) = 0.393x^2 - 0.213x + 8.814$

B. Use the model from part (A), what is the estimate the average length of a cell phone call in July?

$t = 6$

$L(6) = 21.68$ min

Adjust window
X-max: 7
Trace ↑ 6

Domain Restrictions

A function model may have underlying assumptions about how quantities change together. You might need to consider contextual clues that indicate a need to **restrict the domain and/or range** for your model.

EX #5: A custom picture frame is 2 inches wide on all sides. It surrounds a rectangular photo whose length is twice its width, w.

A. Express its total area of the photo and frame A as a function of the width, w

width : $w + 4$
length : $2w + 4$

Area = lw
$A(w) = (2w + 4)(w + 4)$
$= 2w^2 + 12w + 16$

B. What is a reasonable domain and range for this scenario?

Domain : $(0, \infty)$ (width)
Range : $(16, \infty)$ (length)

↑ minimum of the parabola

© 2026 Flamingo Math.com

EX #6: An endangered species is reintroduced into a protected wildlife preserve. The population of the species over time is modeled by the function $P(t) = \frac{10t^2+40t}{t^2+4}$, where t is the time in years since reintroduction and P is the number of individuals in the population.

A. Find $P(4)$. Explain your solution in context.

$P(4) = \frac{10(4)^2+40(4)}{4^2+4} = \frac{320}{20} = 16$

Four years after reintroduction, there are 16 individuals of the species in the preserve.

B. What is the average rate of change from $t = 0$ to $t = 4$? Explain your solution in context.

$AROC = \frac{P(4)-P(0)}{4-0} = \frac{16-0}{4-0} = 4$

Between the time of reintroduction and the 4-year mark, the population increased at an average rate of 4 individuals per year.

C. What is the horizontal asymptote of $P(t)$? What does this value represent in the context of the population?

$y = \frac{10}{1} = 10$

The carrying capacity of the preserve for this species. In the long term, the population is expected to level off at approximately 10 individuals.

EX #7: A photographer notices that the intensity of light, I, from a specific lamp is inversely proportional to the square of the distance, d, from the lamp. At a distance of 4 feet, the light intensity is measured at 50 units.

A. Find the constant of proportionality and write a function to model the intensity I in terms of distance d.

d = ft I = intensity (units)

$I(d) = \frac{k}{d^2} \rightarrow I(4) = 50$

$50 = \frac{k}{4^2} \rightarrow 50 = \frac{k}{16} \rightarrow k = 800$

$I(d) = \frac{800}{d^2}$

B. What would the light intensity be if the photographer moved the lamp to a distance of 10 feet?

$I(10) = \frac{800}{10^2} = 8$ units

C. If the photographer needs an intensity of 32 units for a specific shot, at what distance should the lamp be placed?

$32 = \frac{800}{d^2}$

$32d^2 = 800$

$d^2 = 25$

$d = 5$ feet

© 2026 Flamingo Math.com